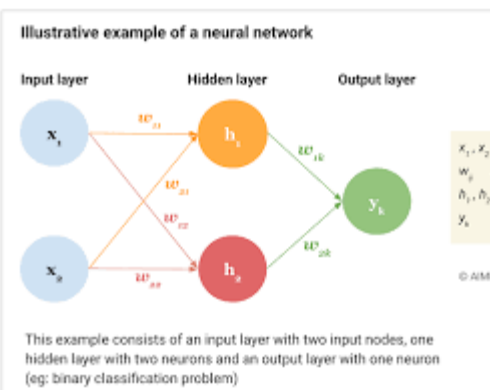
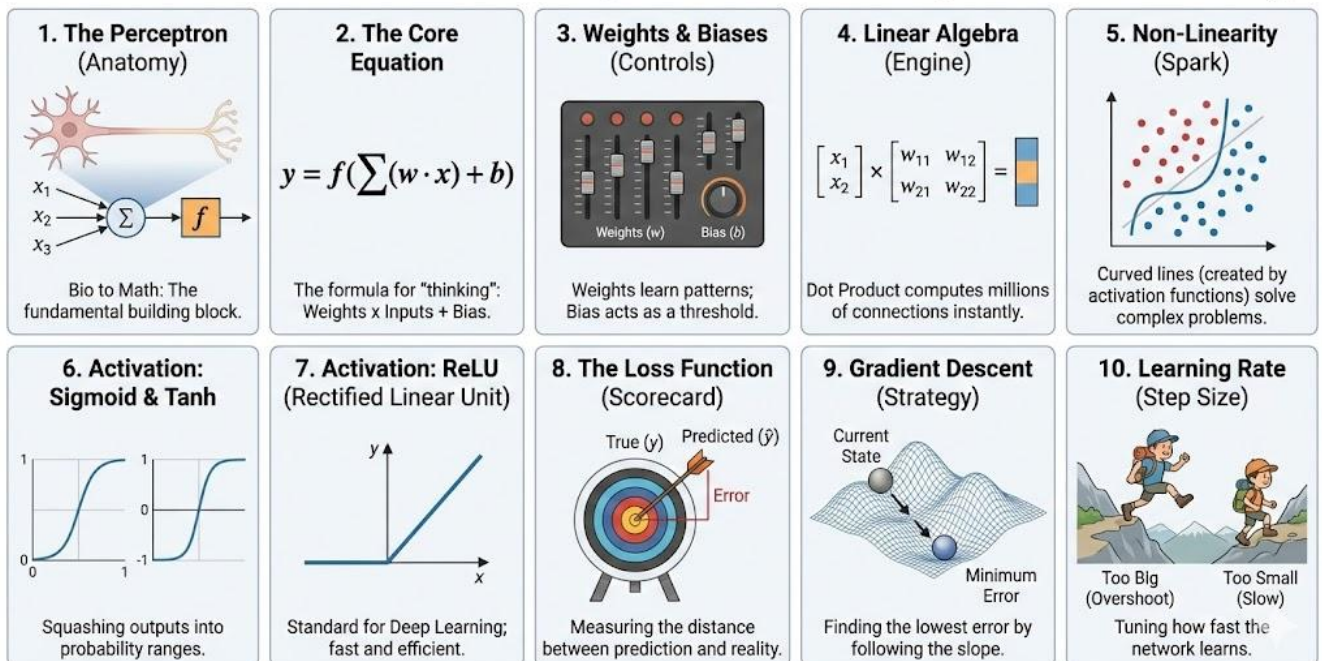


CMT 442 Module 2: The Mathematical Engine of Neural Networks (10-Slide Visual Summary)



COURSE 3: CMT 442 – Mathematical Models & Learning Algorithms

Module 2: Perceptron & Learning Foundations (Expanded)

Slide 1: From Biology to Mathematics

Shift in Focus

- From neurons → equations
- From intuition → computation

Why This Matters

- Governance, bias, explainability depend on math

Slide 2: The Perceptron – Full Breakdown

Formal Definition

- Output = activation(weighted sum + bias)

Key Components

- Inputs (x): features
 - Weights (w): importance
 - Bias (b): threshold
 - Activation: decision function
-

Slide 3: Weights & Biases (Intuition)

Weights

- Control influence
- Learned from data

Bias

- Shifts decision boundary
- Prevents forced zero output

Analogy

- Weights = volume knobs
 - Bias = baseline offset
-

Slide 4: Linear Algebra at the Core

Why Vectors Matter

- Data = vectors
- Model = matrix operations

Dot Product

- Core operation in every layer

Why GPUs Excel

- Massive parallel matrix computation
-

Slide 5: Activation Functions – Why Non-Linearity Matters

Without Activation

- Network collapses into linear regression

With Activation

- Can learn:
 - Curves
 - Boundaries
 - Complex patterns

Slide 6: Activation Functions Compared

Function	Use	Limitation
Sigmoid	Probability	Vanishing gradient
ReLU	Deep nets	Dead neurons
Tanh	Centered output	Still vanishing

Slide 7: Loss Functions – Measuring Error

Purpose

- Quantify “how wrong” the model is

MSE

- Penalizes large errors

Cross-Entropy

- Penalizes confident wrong predictions

Slide 8: Gradient Descent – Optimization Logic

Key Idea

- Move weights to minimize loss

Learning Rate

- Too high → overshoot
- Too low → slow convergence

Why Calculus Matters

- Partial derivatives guide updates
-

Slide 9: Governance Mapping – ISO/IEC 42001

Key Controls

- Data quality
- Bias management
- Model traceability

Audit Need

- Reproducibility of training results
-

Slide 10: NIST AI RMF – Measurement Focus

NIST AI Risk Management Framework

Measure Function

- Accuracy
- Precision
- Recall
- F1-Score

Governance Insight

If performance is not measured, risk cannot be governed

CMT 442 – Mathematical Models & Learning Algorithms

Module 2 Quiz

Questions

Q1. What is the perceptron primarily used to model?

- A. A database query
- B. A biological neuron mathematically
- C. A decision tree
- D. A regression table

Q2. In a perceptron, weights determine:

- A. The number of inputs
- B. The importance of each input
- C. The output label
- D. The learning rate

Q3. What is the role of the bias term in a neuron?

- A. Reducing overfitting
- B. Increasing computation speed

- C. Shifting the activation threshold
- D. Normalizing input data

Q4. Which mathematical operation is fundamental to neural networks?

- A. Matrix inversion
- B. Dot product
- C. Sorting
- D. Hashing

Q5. Why are activation functions necessary in neural networks?

- A. To reduce computation
- B. To add non-linearity
- C. To normalize outputs
- D. To prevent overfitting

Q6. Which activation function is most commonly used in deep networks?

- A. Sigmoid
- B. Tanh
- C. Step function
- D. ReLU

Q7. Which loss function is commonly used for classification problems?

- A. Mean Squared Error
- B. Absolute Error
- C. Cross-Entropy
- D. Euclidean Distance

Q8. What does gradient descent aim to minimize?

- A. Weights
- B. Bias
- C. Loss function
- D. Learning rate

Q9. A very high learning rate in gradient descent may cause:

- A. Faster convergence always
- B. Overshooting the optimal solution
- C. No weight updates
- D. Better accuracy

Q10. Why is model performance measurement critical for AI governance?

- A. To improve hardware efficiency
- B. To reduce training time
- C. To support auditability and risk management
- D. To eliminate bias completely

Answers

1. B
2. B

3. **C**

4. **B**

5. **B**

6. **D**

7. **C**

8. **C**

9. **B**

10. **C**

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